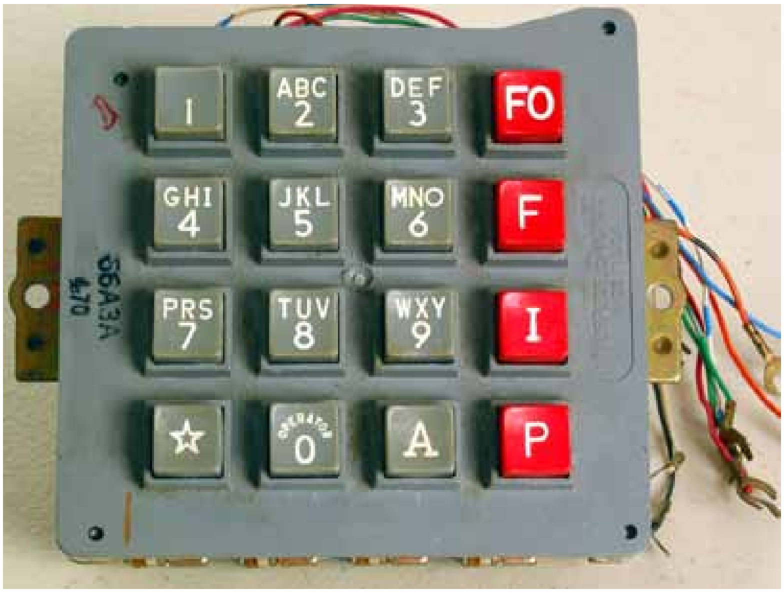
DTMF: The Evergreen Retro Tech

This article looks at how DTMF technology has evolved over the years, and can still be of use



(Credit: https://en.wikipedia.org)

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In the days of analogue telephone communication, the most common type of cable that was used for connectivity was the unshielded twisted pair (UTP). Since it did not have a physical shield to block any interference, the amount of interference was a bit high. However, it was not too noisy and was good enough to transmit voice. But as the days passed and the number of people using telephones increased, the need for an automated telephone exchange increased too. It was in January 1958 that G. Goertzel published a paper called 'An Algorithm for the Evaluation of Finite Trigonometric Series,' in the *American Mathematical Monthly.* This became the basis of dual tone multiple frequency (DTMF) signalling. It allowed us to encode numeric values as superpositions of two co-prime sine waves in such a way that they became noise-resistant.

On November 18, 1963, Bell Systems introduced phones with buttons instead of rotary dials. DTMF assigned a special tone to each button pressed on the telephone device. As a number was dialled, each button that was pressed gave a standard audio signal. The telephone exchange could decode these signals as DTMF to find out what number the user had typed in.

Binary was defined by 'on' and 'off' or '1' and '0'. DTMF technology worked by having the handset generate tones at specific frequencies and playing them over the phone line when a button was pressed on the keypad. It could be treated like a device to switch up to eight appliances. Equipment at the other end of the phone line listened to the specific sounds and decoded them into commands.

The Goertzel algorithm is a digital signal processing (DSP) method for quickly evaluating each discrete Fourier transform term (DFT). It is beneficial in some real-world situations, such as when recognising the DTMF tones emitted by the pushbuttons on a classic analogue telephone's keypad. The spectral data can be quickly and efficiently extracted from an input signal using this approach. In order to efficiently compute DFT values, it essentially makes use of two-pole IIR type filters. As a result, it is a recursive structure that always processes one incoming sample at a time, as opposed to the DFT (or FFT), which must first have a block of data to begin processing.

Now, let's break this down into several simple steps.

A DTMF signal consists of two sine